Realtime app con Blazor ed Azure

Sabato 24 settembre 2022

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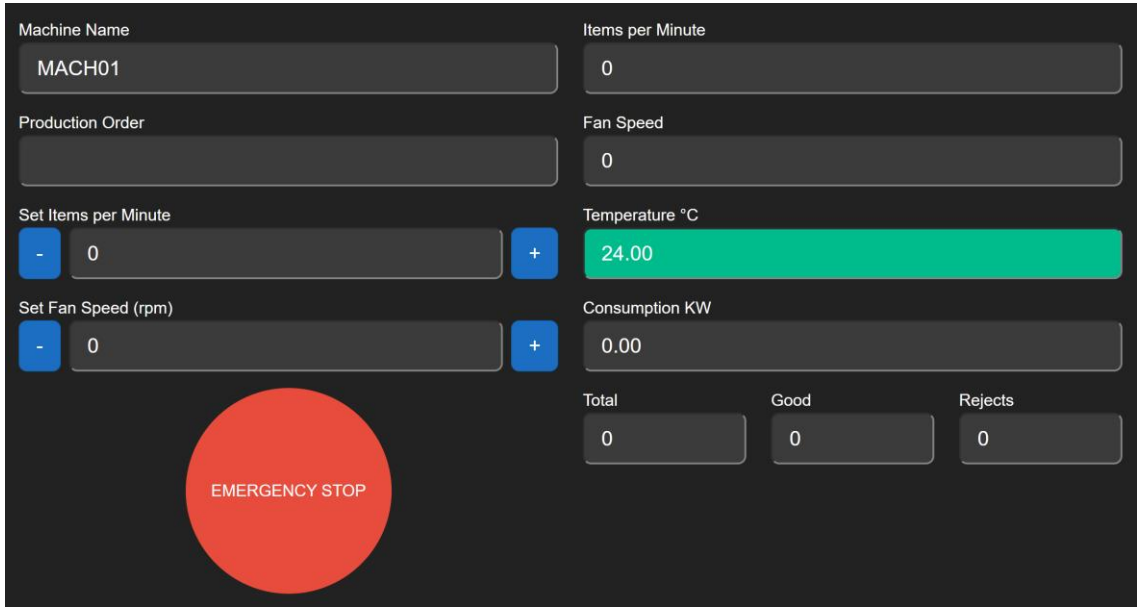
A Real Life Story (maybe)

C’era una volta Itemfactory S.R.L. …

- Azienda specializzata nella produzione di oggetti
- Linee di produzione già connesse ad un HMI
- I dati arrivano già in Cloud

- Vuole un applicativo MES per la gestione degli ordini di produzione
  - Lo vuole Desktop, Mobile e Web
  - Vuole che si aggiorni in tempo reale
  - Vuole che costi poco



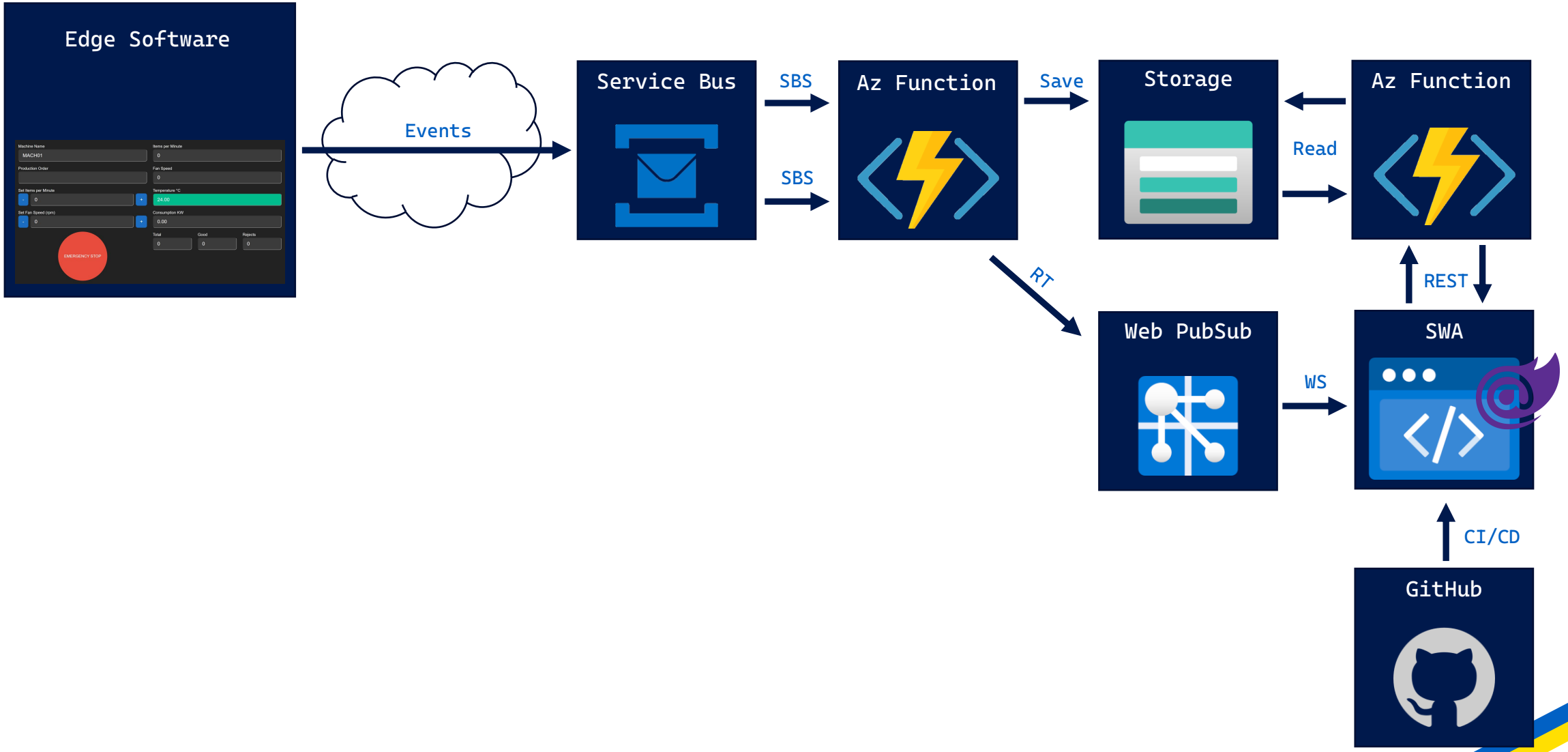
HMI ⬆️

(facciamo finta che lo sia)

The Recipe

Cross Platform	Hosting	UI Framework	Data Storage	Data Movement + Data Access	Realtime Messaging System	Code Repo + CI/CD
Progressive Web App	Azure Static WebApp	Blazor	Azure Storage Account	Azure Functions	Azure Web PubSub	GitHub
Free	Free Tier	Free	Pay per Use	Consumption Plan	Free Tier	Free

The Architecture Schema



Blazor e PWA

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Cos’è Blazor

Blazor è un framework per la creazione di interfacce web interattive client-side utilizzando .NET.

Cosa possiamo fare?

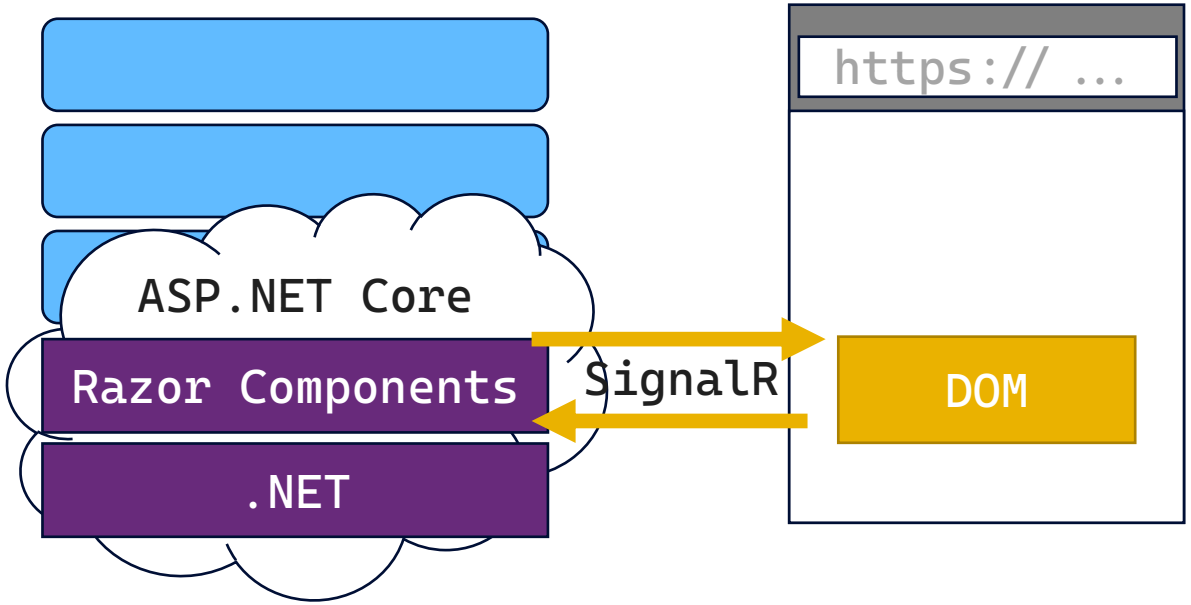
- Creare interface web utilizzando C# invece che JS*
- Condividere logica tra server e client
- Utilizzare librerie .NET anche lato client**

* Non è vero, JS rimane per molte cose l’unica strada percorribile

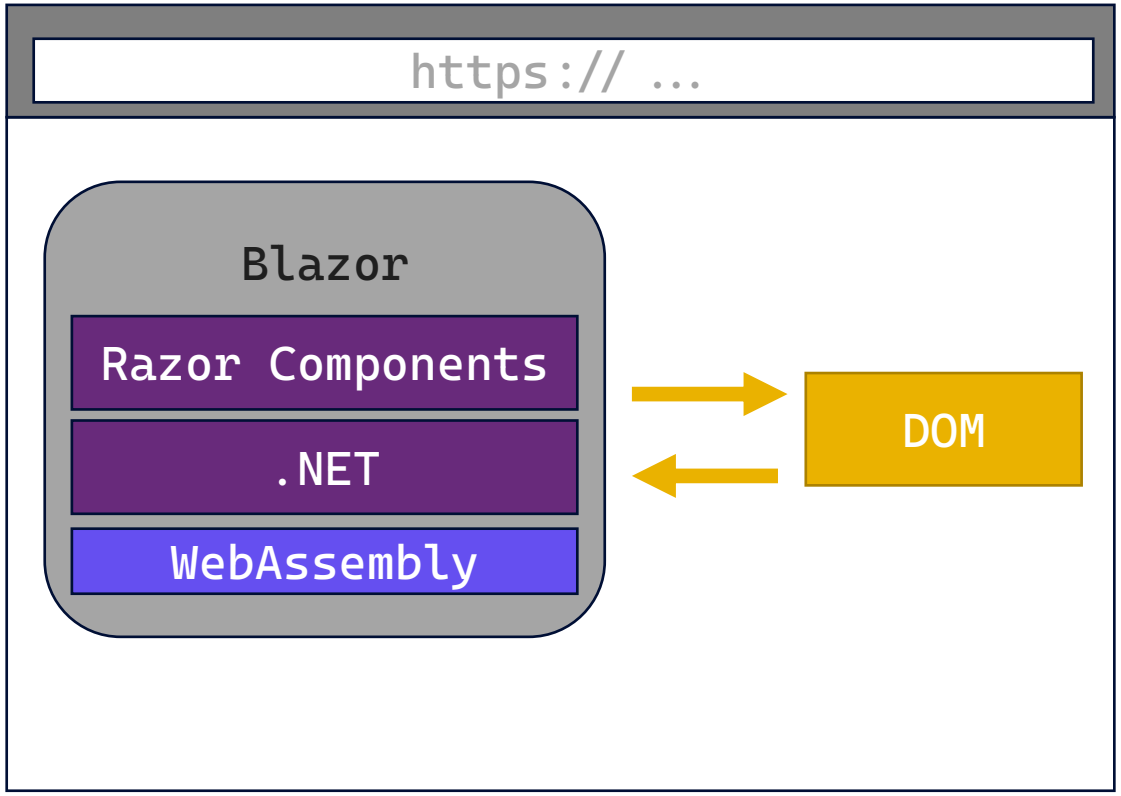
** Con alcune restrizioni.

Blazor Server vs WebAssembly

Blazor Server



Blazor WebAssembly (WASM)



Cos’è una PWA?

È un applicativo web installabile.

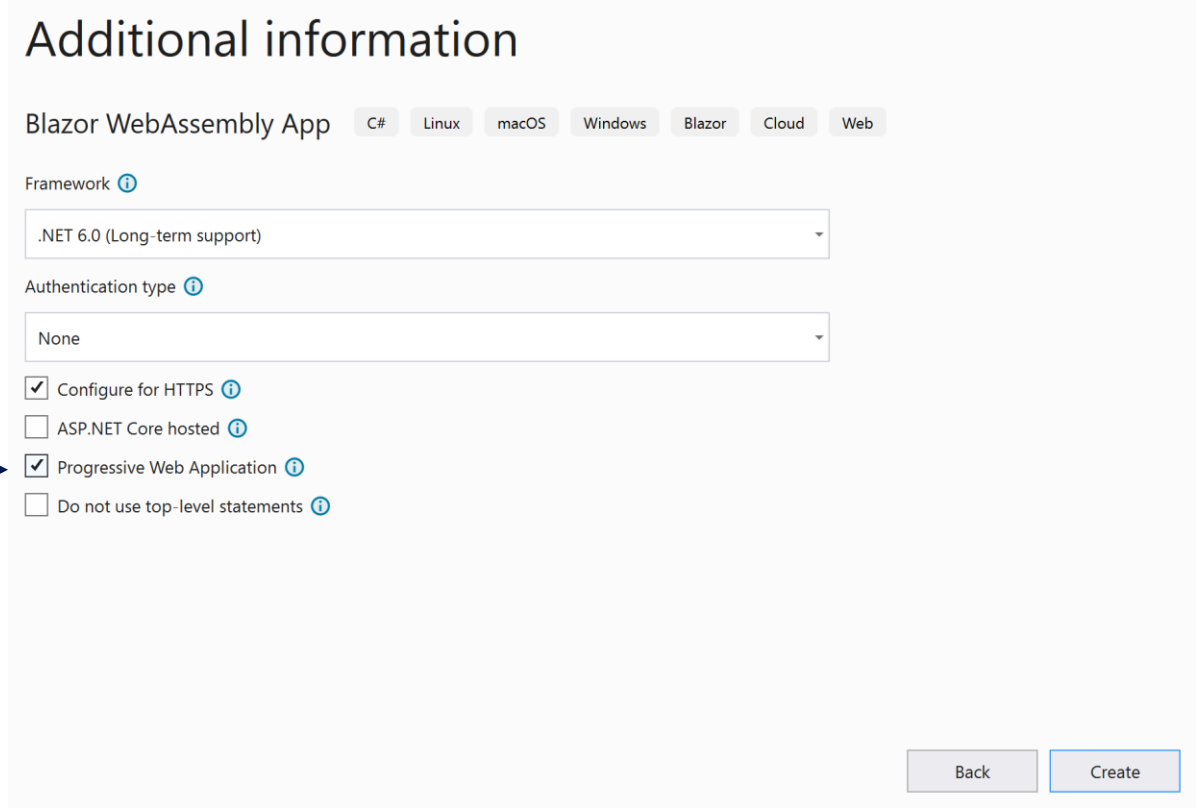
Deve soddisfare tutti i seguenti requisiti:

- Contenere un manifest.json.
- Servito in HTTPS
- Contenere un’icona per la Shortcut
- Deve registrare un service worker
  richiesto da tutti i browser Chromium-based

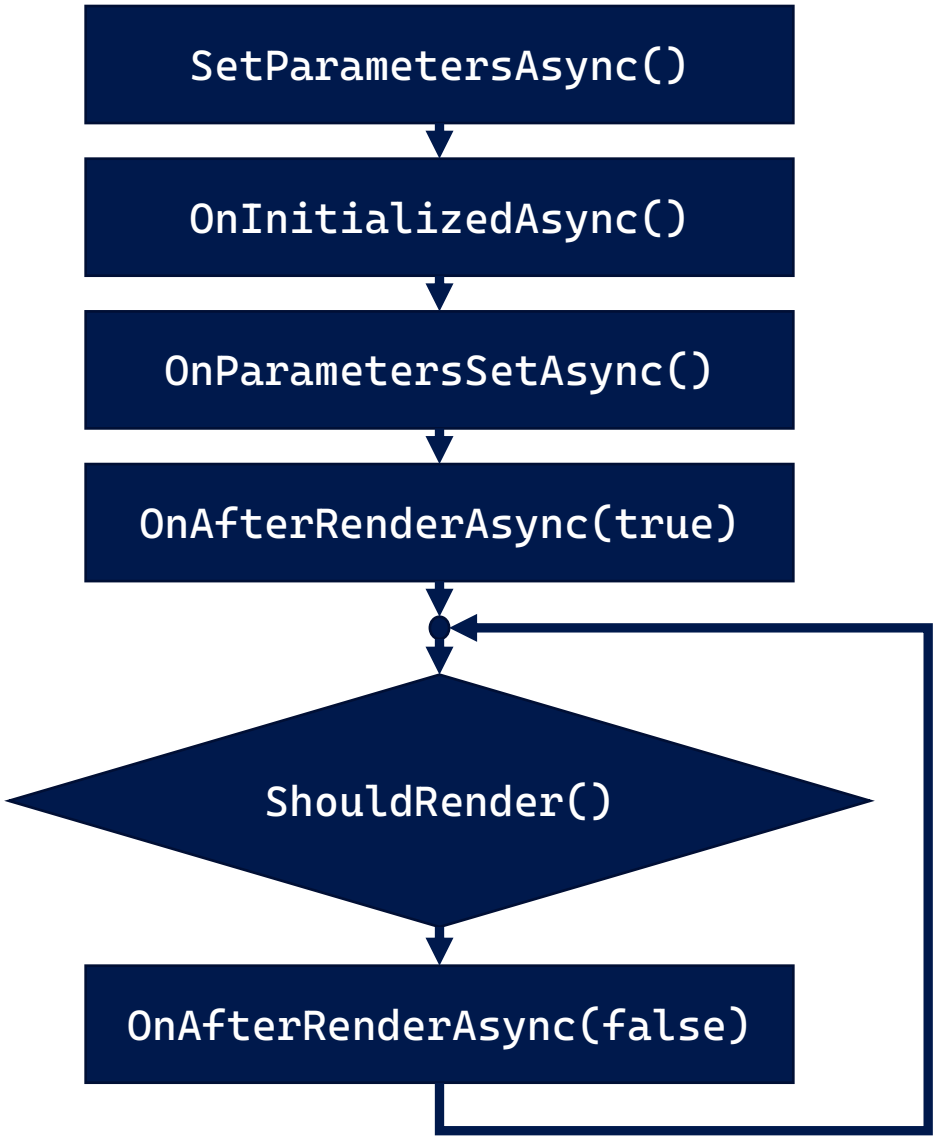
Come posso creare una PWA con Blazor?

Opzioni:

1. Creo un’app AspNetCore e aggiungo gli assets per renderlo una PWA
2. dotnet new blazorwasm -o AppName --pwa
3. Così:



Razor Components Lifecycle



Special Guest

DisposeAsync()

Dispose() e DisposeAsync() sono invocati all'unload del componente se il componente implementa IDisposable o IAsyncDisposable

Blazor Page Structure

```
@page "/"
@inject FactorySensors sensors
@implements IDisposable

<h1>Hello World</h1>

@code {

    protected override Task OnAfterRenderAsync(bool firstRender)
    {
        return base.OnAfterRenderAsync(firstRender);
    }

    public void Dispose() { }

}
```

Azure Web PubSub

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Azure Web PubSub

Servizio gestito che supporta WebSockets serverless.

Pensato per applicazioni che richiedono aggiornamenti in tempo reale

- Aggiornamenti ad alta frequenza
- Live dashboards e monitoring
- Cross-platform live chat
- Real-time location on map
- Real-time targeted ads
- Collaborative apps
- Notifiche Push
- Real-time broadcasting
- IoT e dispositivi connessi
- Automazione

SignalR vs Azure Web PubSub

- Stai già usando SignalR
- Esiste un client SignalR nel linguaggio scelto
  - Clients supportati da MS:

- Hai necessità di più protocolli di trasporto
  - WebSocket
  - Server sent events
  - Long polling

- Hai bisogno solo di WebSocket
- Subprotocollo custom over WebSocket
  - MQTT over WebSocket
  - AMQP over WebSocket
- È sufficiente un'infrastruttura «lite» per la tua applicazione

* Azure Web PubSub è solo PaaS

Azure Web PubSub Pricing

Pricing tier	Free For individual dev/test	Standard For production workloads	Premium(Preview) For production workloads with more supported features
Features			
Connections	Up to 20 connections	1,000 connections/unit	1,000 connections/unit
Included Messages	20,000/Day	1,000,000/Unit/Day	1,000,000/Unit/Day
Additional Messages	-	Unlimited	Unlimited
SLA	-	99.9%	99.95%
SSL	✓	✓	✓
Autoscale ⓘ	-	-	✓
Availability Zone ⓘ	-	-	✓ (In supported regions)
Custom Domains	-	-	✓
Pricing			
Estimated Price	-	41.41 EUR/Month/Unit	51.44 EUR/Month/Unit
Additional Message Costs ⓘ	-	0.84 EUR per million messages	0.84 EUR per million messages

DEMO – Azure Web PubSub

Sending Data with Azure Web PubSub

- ServiceClient C# Azure.Messaging.WebPubSub
- AzFunction Bindings Microsoft.Azure.WebJobs.Extensions.WebPubSub

```
[FunctionName(nameof(EventNotifier))]
public static async Task NotifyEventAsync(
    [ServiceBusTrigger("%TopicName%", "%SubscriptionName%", Connection = "ServiceBusConnectionString")] string rawpayload
    , [WebPubSub(Hub = "alarms", Connection = "WebPubSubConnectionString")] IAsyncCollector<WebPubSubAction> alarms
    , ILogger log
)
{
    var payload = Payloads.DeserializePayload(rawpayload);

    if (payload is null)
        return;

    if (payload is AlarmPayload)
    {
        await alarms.AddAsync(WebPubSubAction.CreateSendToAllAction(rawpayload));
        await alarms.FlushAsync();
    }
}
```

Get AccessUri for Azure Web PubSub

- Tramite il portale Azure

- ServiceClient C#

```
var clientAccessUri = await serviceClient.GetClientAccessUriAsync();
```

- AzFunction Bindings

```
[FunctionName(nameof(NegotiateAlarms))]
public static WebPubSubConnection NegotiateAlarms(
[HttpTrigger(AuthorizationLevel.Anonymous, "get", "post", Route = "alarms/negotiate")] HttpRequest req,
[WebPubSubConnection(Hub = "alarms", Connection = "WebPubSubConnectionString")] WebPubSubConnection connection)
{
    return connection;
}
```

DEMO – Notifiche in pagina

Azure Static WebApp

Cos’è Azure Static WebApp

È un servizio che consente di creare applicazioni Web statiche.

La pubblicazione dei nuovi contenuti avviene tramite un sistema di CI/CD.

Si integra nativamente con GitHub e con Azure DevOps.

Az Blob Static Website vs Az Static WebApp

- ✔ SSL
- 😟 CI/CD non gestito
- ❌ No Multiregion
- ✔ Custom Domain
- ❌ No HTTPS con Custom Domains

- ✔ SSL
- ✔ CI/CD integrate
- ✔ Globalmente distribuito
- ✔ Custom Domain
- ✔ HTTPS con Custom Domain
- ✔ API proxy integrato

Azure Static WebApp – Api

Free Tier

- Azure Functions

Standard Tier

- Azure Functions
- Azure Api Management
- Azure AppService
- Azure Container Apps

Azure Static WebApp Pricing

Plan/Features	**Free** For hobby or personal projects	**Standard** For general purpose production apps
Price	Free	7.59 EUR/per app/month
Included bandwidth	100GB per subscription	100GB per subscription
Bandwidth overage	Free	0.17 EUR per GB per subscription
Custom domains	2 per app	5 per app
SSL certificates	Free	Free
Custom authentication	-	✓
Private endpoints	-	✓
Max app size	250MB	500MB
Staging environments	3	10
Azure Functions	Managed	Managed or Bring your own
Enterprise-grade edge	-	14.77 EUR/per app/month

DEMO – Azure Static WebApp

Creare una Azure Static WebApp

Basics Tags Review + create

App Service Static Web Apps is a streamlined, highly efficient solution to take your static app from source code to global high availability. Pre-rendered content is distributed globally with no web servers required. Learn more

Project Details

Select a subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Nicola Paro

Resource Group * rg-blazormes

Create new

Static Web App details

Name * mynewstaticwebapp

Hosting plan

The hosting plan dictates your bandwidth, custom domain, storage, and other available features. Compare plans

Plan type

- (●) Free: For hobby or personal projects
- ( ) Standard: For general purpose production apps

Azure Functions and staging details

Region for Azure Functions API and staging environments * West Europe

Deployment details

Source

- (●) GitHub
- ( ) Azure DevOps
- ( ) Other

GitHub account nicolaparo

Change account

Creare una Azure Static WebApp

If you can't find an organization or repository, you might need to enable additional permissions on GitHub.

Organization * nicolaparo

Repository * blazormesdemo

Branch * main

Build Details

Enter values to create a GitHub Actions workflow file for build and release. You can modify the workflow file later in your GitHub repository.

Build Presets Blazor

These fields will reflect the app type's default project structure. Change the values to suit your app.

App location * Client

Api location Api

Output location wwwroot

Preview workflow file

Review + create | < Previous | Next : Tags >

Framework

- Angular
- Blazor
- Next.js
- Nuxt.js
- React
- Svelte
- Vue.js

Static site generator

- Gatsby
- Hugo
- VuePress

Other

- Custom

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Creare una Azure Static WebApp

https://victorious-tree-017109303.1.azurestaticapps.net

Q & A

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Thanks!

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